

FRIDGEGUARD SYSTEM

Technical Documentation & Code Breakdown

1. System Overview

FridgeGuard is an IoT monitoring solution designed to detect food spoilage in real-time. By utilizing an **MQ-135 Gas Sensor** and a **DHT22 Temperature/Humidity Sensor**, the system identifies gas fluctuations caused by decaying organic matter and alerts the user through an OLED display and a smart buzzer system.

2. Hardware Assignment

Component	GPIO Pin	Interface
OLED SCL	GP1	I2C (Bus 0)
OLED SDA	GP0	I2C (Bus 0)
DHT22 Sensor	GP15	Digital Input
MQ-135 Sensor	GP26	Analog Input (ADC)
Buzzer	GP16	Digital Output

3. Code Explanation

Step 1: Library Initialization

```
from machine import Pin, I2C, ADC
import dht, time, ssd1306
```

Pin/ADC: Controls GPIO and reads analog gas data.

I2C/ssd1306: Handles communication with the OLED display.

dht: Specifically designed to interface with DHT11/22 sensors.

Step 2: Display & Sensor Setup

```
i2c = I2C(0, scl=Pin(1), sda=Pin(0))
oled = ssd1306.SSD1306_I2C(128, 64, i2c)
dht_sensor = dht.DHT22(Pin(15))
mq135 = ADC(26)
buzzer = Pin(16, Pin.OUT)
```

Configures the I2C bus at **Bus 0** and initializes the 128x64 display. Sets up the environmental sensors and the alert buzzer on their respective pins.

Step 3: Auto-Calibration (The Baseline)

Calibration Logic: To handle varying background pollution (e.g., Delhi/NCR air), the system takes 10 initial readings and calculates an average.

```
baseline = 0
for i in range(10):
    baseline += mq135.read_u16()
    time.sleep(0.2)
baseline /= 10
```

This prevents false alarms by ensuring only **changes** from the current environment are monitored.

Step 4: Real-Time Decision Logic

The system calculates the difference ($\text{diff} = \text{gas} - \text{baseline}$) to determine spoilage status:

- **SPOILED** **Diff > 15,000:** Critical gas levels detected.
- **WARNING** **Diff > 8,000:** Moderate increase; inspection recommended.

- **SAFE** Other: Air quality within normal parameters.

Step 5: Smart Alert System

The buzzer only activates when the status shifts to **SPOILED** or after a 10-second cooldown. This prevents redundant, "annoying" noises while maintaining safety awareness.

4. User Interface Model

The OLED display updates every 2 seconds provided the following information:

```
[ FridgeGuard ]  
Temp: 24C  
Gas: 18500  
Diff: 4200  
SAFE
```

5. Robustness & Error Handling

A global `try . . . except` block ensures that if a sensor wire is disconnected or a reading fails, the **buzzer is silenced** and the system attempts a retry instead of crashing.